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Project: /Volumes/projectDrive/projects/webDev/ai/langchain/langchain_tutorial Python: /Volumes/projectDrive/projects/webDev/ai/langchain/langchain_tutorial/.venv/bin/python Running 5 script(s) in lesson_15/

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=====
[1/5] Running: howToA-B.py =====
Running A/B test with 2 queries... Testing 2 variants: ['baseline', 'optimized']
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=== A/B Test Report ===

baseline: queries: 2, success: 100.00% response_time: mean=346.201 (?288.467) relevancy_score: mean=0.550 (?0.198) cost: mean=0.000 (?0.000) user_satisfaction: mean=0.544 (?0.047) optimized: queries: 2, success: 100.00% response_time: mean=135.187 (?8.259) relevancy_score: mean=0.550 (?0.198) cost: mean=0.000 (?0.000) user_satisfaction: mean=0.491 (?0.172)

Recommendations: - Variant 'optimized' shows the best overall performance

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[2/5] Running: howToCreateCustomMetrics.py =====
Hallucination Score: 0.250 Completeness Score: 0.400 Response Time Score: 1.000
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[3/5] Running: howToUse.py =====
For more information check: https://developer.mozilla.org/en-US/docs/Web/HTTP/Status/402
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You have depleted your monthly included credits. Purchase pre-paid credits to continue using Inference Providers. Alternatively, subscribe to PRO to get 20x more included usage.)

For more information check: <https://developer.mozilla.org/en-US/docs/Web/HTTP/Status/402>

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Bad request:)

=== RAG System Evaluation Report ===

Overall Scores: context_precision: 1.000 context_recall: 0.833 answer_relevancy: 0.652 faithfulness: 1.000 answer_correctness: nan

Metric Analysis:

context_precision: Mean: 1.000 (?0.000) Range: [1.000, 1.000] Median: 1.000

context_recall: Mean: 0.833 (?0.236) Range: [0.667, 1.000] Median: 0.833

answer_relevancy: Mean: 0.652 (?nan) Range: [0.652, 0.652] Median: 0.652

faithfulness: Mean: 1.000 (?nan) Range: [1.000, 1.000] Median: 1.000

answer_correctness: Mean: nan (?nan) Range: [nan, nan] Median: nan

Areas for Improvement: [!] answer_correctness: failed (no score) [!] answer_relevancy: 0.652

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[4/5] Running: monitorProduction.py =====
ERROR: Query hf_0 failed: Client error '402 Payment Required' for url 'https://router.huggingface.co/v1/chat/completions'
(Request ID: Root=1-6a0dbb79-3e94134e2fe8a6136bc2956a;52bb45c4-158a-4652-b077-314a928dee5c) For
more information check: https://developer.mozilla.org/en-US/docs/Web/HTTP/Status/402
```

You have depleted your monthly included credits. Purchase pre-paid credits to continue using Inference Providers. Alternatively, subscribe to PRO to get 20x more included usage. WARNING: Alert: error_rate is 1.000 (above threshold 0.05) WARNING: Alert: relevancy_score is 0.000 (below threshold 0.7) ERROR: Query hf_1 failed: Client error '402 Payment Required' for url 'https://router.huggingface.co/v1/chat/completions' (Request ID: Root=1-6a0dbb79-2d7f5c6b2cfdd78e2d1f4f15;fbc95cb4-ccde-445d-b627-6e1c20bd6b8b) For more information check: <https://developer.mozilla.org/en-US/docs/Web/HTTP/Status/402>

You have depleted your monthly included credits. Purchase pre-paid credits to continue using Inference Providers. Alternatively, subscribe to PRO to get 20x more included usage. WARNING: Alert: error_rate is 1.000 (above threshold 0.05) WARNING: Alert: relevancy_score is 0.000 (below threshold 0.7) WARNING: Alert: error_rate is 0.667 (above threshold 0.05) WARNING: Alert: relevancy_score is 0.302 (below threshold 0.7) WARNING: Alert: error_rate is 0.500 (above threshold 0.05) WARNING: Alert: relevancy_score is 0.456 (below threshold 0.7) ERROR: Query sim_2 failed: Timeout WARNING: Alert: error_rate is 0.600 (above threshold 0.05) WARNING: Alert: relevancy_score is 0.553 (below threshold 0.7) WARNING: Alert: error_rate is 0.500 (above threshold 0.05) WARNING: Alert: relevancy_score is 0.587 (below threshold 0.7) WARNING: Alert: error_rate is 0.429 (above threshold 0.05) WARNING: Alert: relevancy_score is 0.638 (below threshold 0.7) WARNING: Alert: error_rate is 0.375 (above threshold 0.05) WARNING: Alert: relevancy_score is 0.656 (below threshold 0.7) WARNING: Alert: error_rate is 0.333 (above threshold 0.05) WARNING: Alert: relevancy_score is 0.676 (below threshold 0.7) WARNING: Alert: error_rate is 0.300 (above threshold 0.05) WARNING: Alert: relevancy_score is 0.695 (below threshold 0.7) WARNING: Alert: error_rate is 0.273 (above threshold 0.05) WARNING: Alert: error_rate is 0.250 (above threshold 0.05) Recording live HF queries... === RAG System Dashboard (Hugging Face) === Model: meta-llama/Llama-3.1-8B-Instruct (provider: auto) Embeddings: sentence-transformers/all-MiniLM-L6-v2 (hf-inference) Health Score: 61.6/100

Current Metrics (5 min): error_rate: 0.250 p50_latency_ms: 1146.198 p95_latency_ms: 1635.572 p99_latency_ms: 1636.967 relevancy_score: 0.716 cost: 0.000 cost_per_query: 0.000

Recent Alerts: [09:51:43] Alert: relevancy_score is 0.676 (below threshold 0.7) [09:51:43] Alert: error_rate is 0.300 (above threshold 0.05) [09:51:43] Alert: relevancy_score is 0.695 (below threshold 0.7) [09:51:43] Alert: error_rate is 0.273 (above threshold 0.05) [09:51:43] Alert: error_rate is 0.250 (above threshold 0.05)

Trends (newer half vs older half of session when run is short): error_rate: v 100.0% p50_latency_ms: ^ 18.7% p95_latency_ms: v 14.6% p99_latency_ms: v 14.4% relevancy_score: ^ 32.6% cost: ^ 42.1% cost_per_query: ^ 65.8%

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[5/5] Running: optimizeRagCost.py =====
Model: meta-llama/Llama-3.1-8B-Instruct Embeddings: sentence-transformers/all-MiniLM-L6-v2 (hf-inference) Monthly cost (estimate): \$19.88 Cost per query (estimate): \$0.0004

Cost breakdown: embedding: \$0.38 retrieval: \$0.00 generation: \$19.50

Tips: - Lower retrieval_k to 3 to cut input tokens to the chat model. - Use ~500-token chunks to reduce context size per query. - Chat uses Inference Providers (auto); embeddings use hf-inference - track both quotas.

Optimization suggestions: - Reduce chunk size to 500 tokens: ~20% potential savings - Reduce retrieval_k to 3: ~40% potential savings - Cache HF responses for common queries (disk or Redis): ~60% potential savings - Run embeddings on hf-inference; keep chat on auto only when needed: ~15% potential savings

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Summary =====
howToA-B.py: OK howToCreateCustomMetrics.py: OK howToUse.py: OK monitorProduction.py: OK
optimizeRagCost.py: OK

All scripts completed successfully. Finished: 2026-05-20T09:51:44